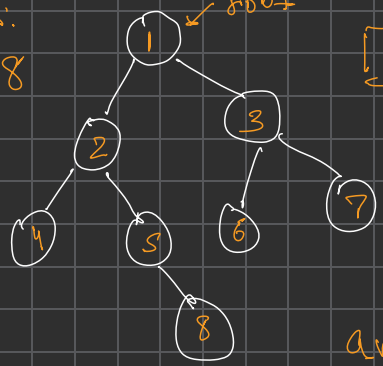
Level Order Traversal ⇒ O(n)

vector<vector<int>>ans;

temp = 8



result

T.Compl

```
while(!q.empty())
{
    int len = q.size()
    vector<int> result;
    while(len--)
    {
        Node * temp = q.front();
        q.pop();
        result.push_back(temp->data)
        if(temp->left)
        {   q.push(temp->left)
        }
        if(temp->right)
        {   q.push(temp->right);
        }
    }
    ans.push_back(result);
}
```

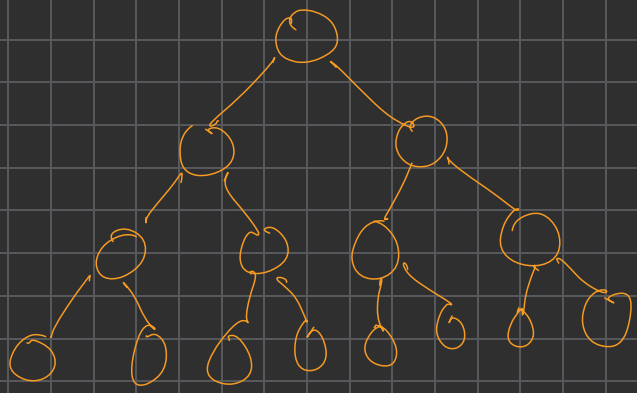
len = 1

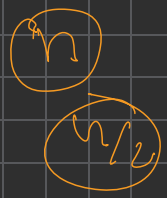
ans = 1

2 3

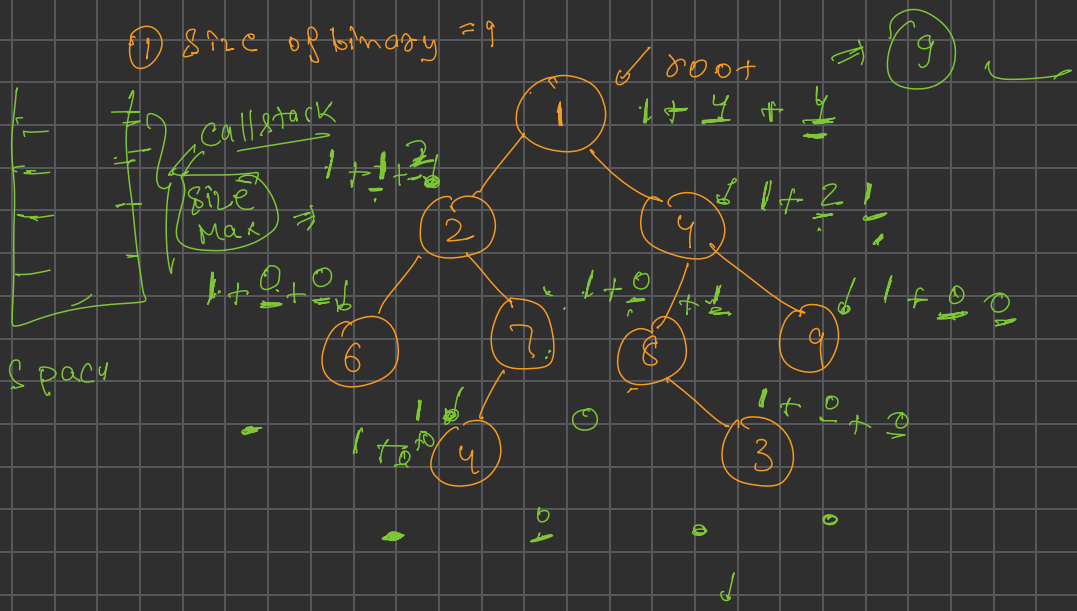
4 5 6 7

8




n
n/2

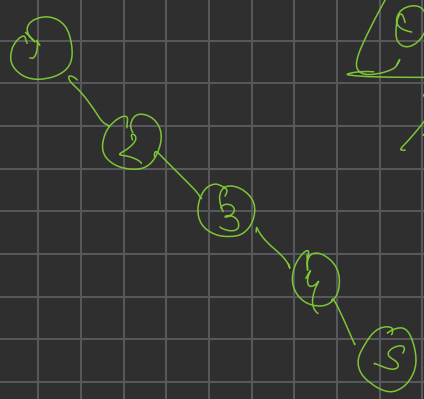
(1) size of binary = 9

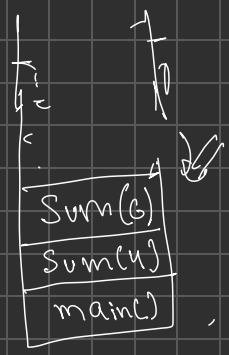


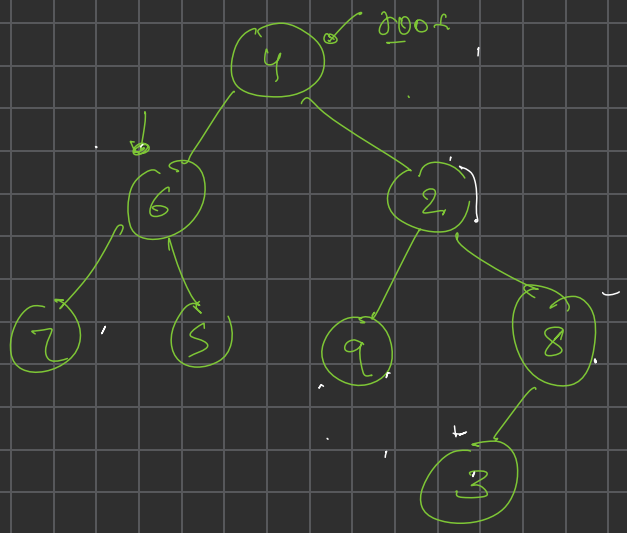

```
int size(Node *root)
{
    if(!root)
        return 0;

    return 1 + size(root->left) + size(root->right)
}
```

$\boxed{O(n)}$

Binary

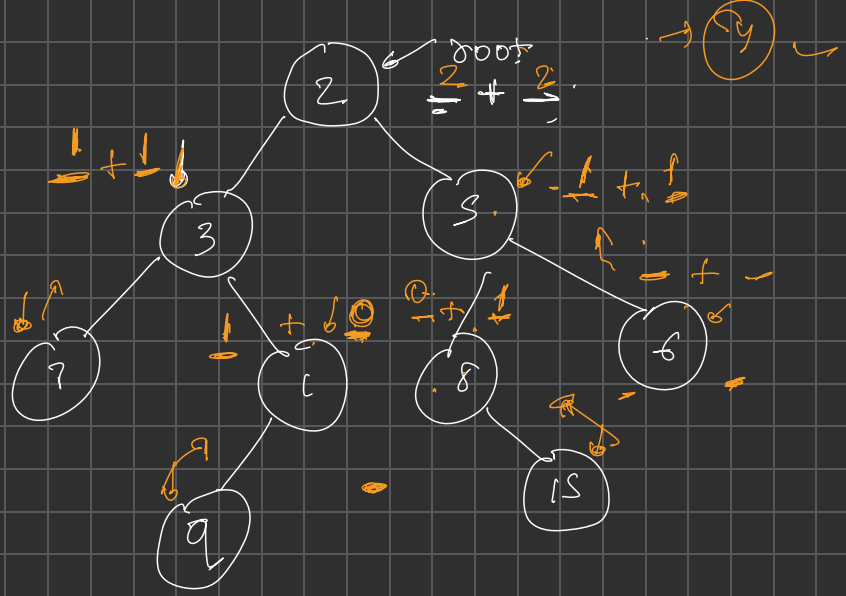




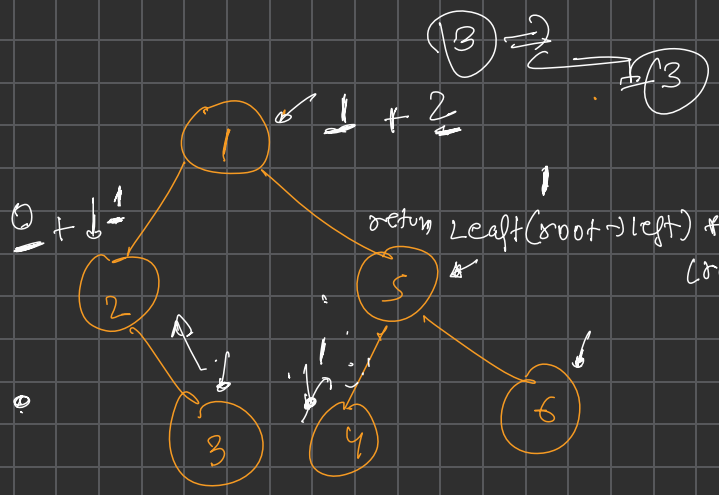


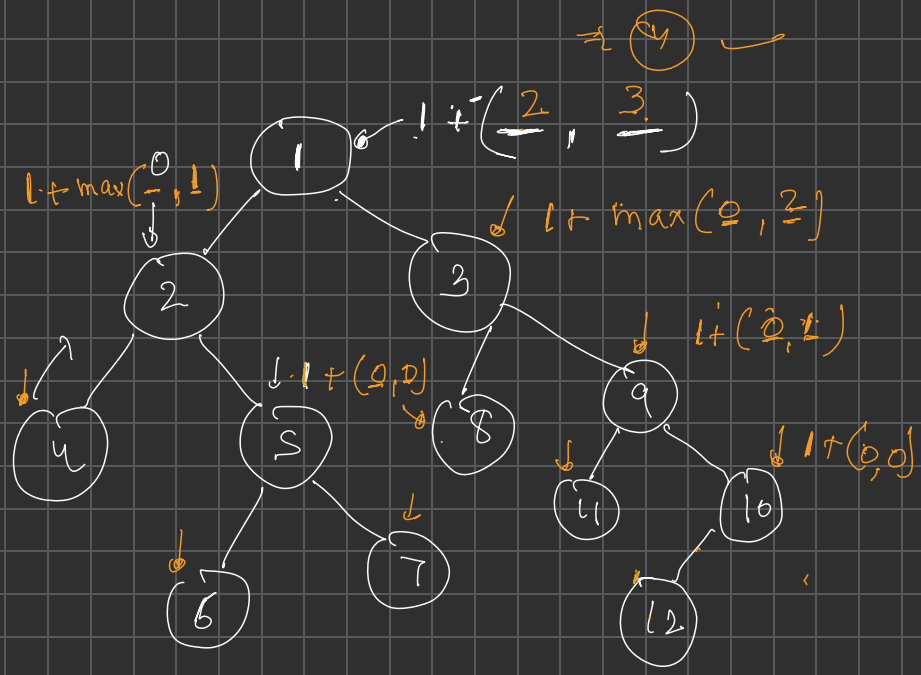

```
int main()
{
  sum(root);

}
```

```
int sum(Node *root)
{       if(!root)
            return 0;

return root->data + sum(root->left) + sum(root->right)

}
```

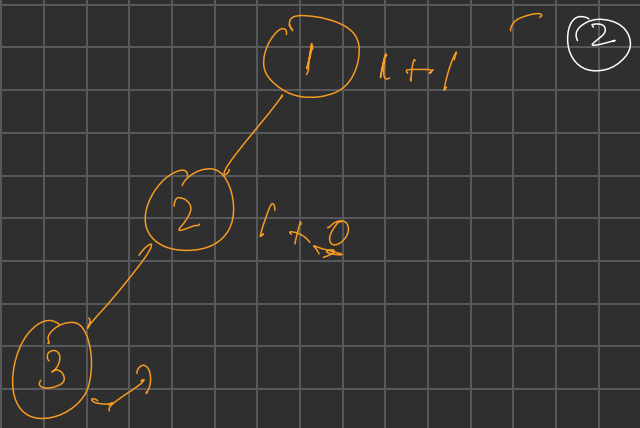


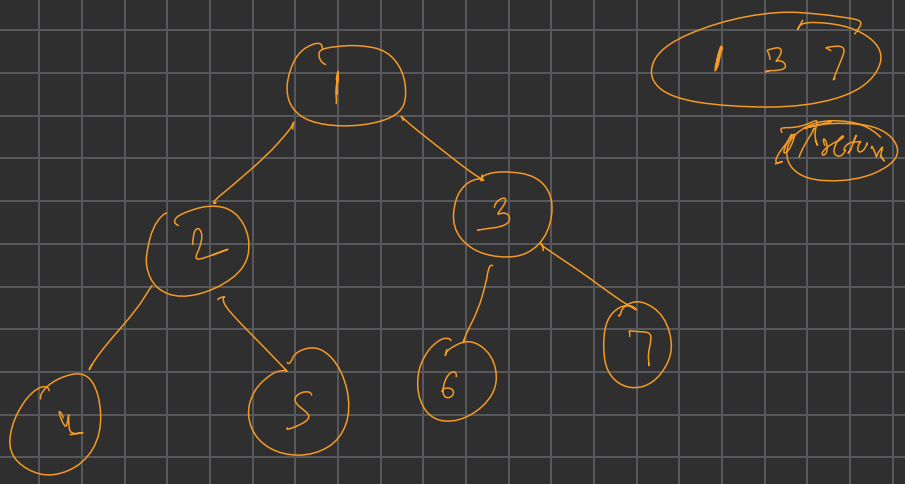

```
int count(Node * root)
{
    if(!root)
        return 0;

    // leaf
    if(!root->left && !root->right)
        return 1;

    return count(root->left) + count(root->right)
```

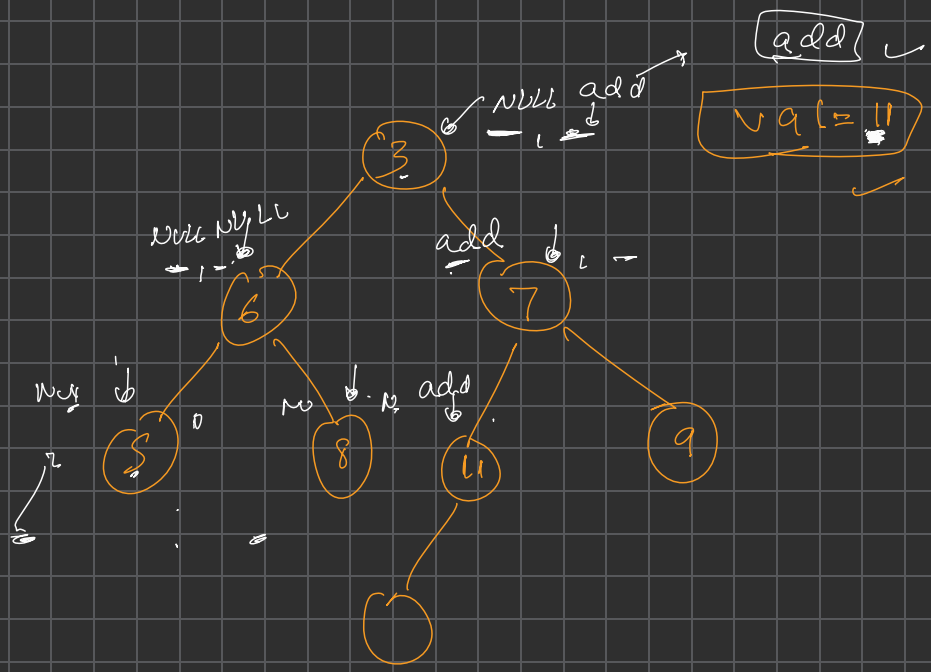

B
3
1 + 2
0 + 1
return Leaf(root->left) + Leaf(root->right)
1
2
5
3
4
6



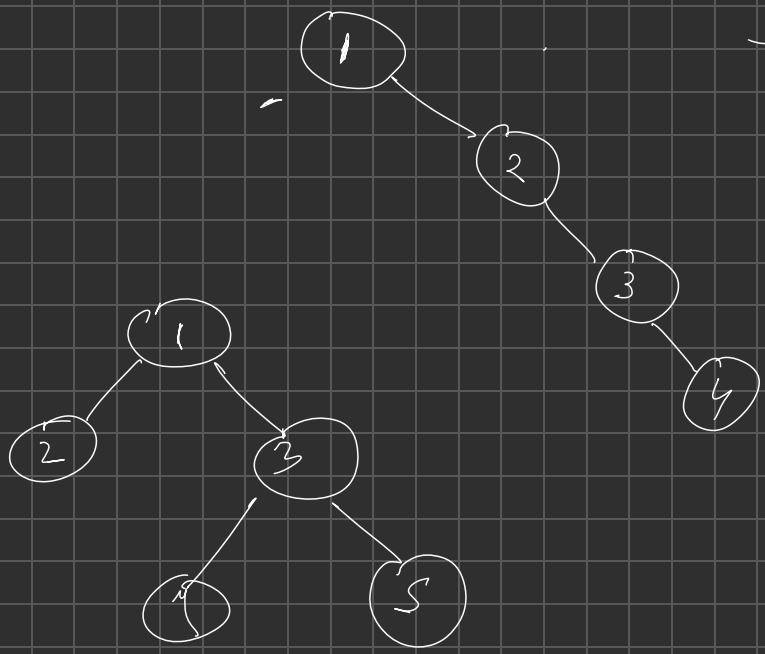

```
int Height( Node *root)
{
    if ( !root)
        retur 0;
    if ( !root->left && !root->right)
        return 0;

return 1 + max(Height(root->left), Height(root->right))
}
```


1
1+1
2
2
1+0
3
1


1
2
3
4
5
6
7
1 3 7
return




```
Node* Find (Node *root, val)
{
    if (!root)
        return NULL

    Node* LeftSide = Find(root->left, val)
        if (LeftSide)
            return LeftSide;

    return Find(root->right, val)
}
```


1
2
3
4
1
2
3
4
5